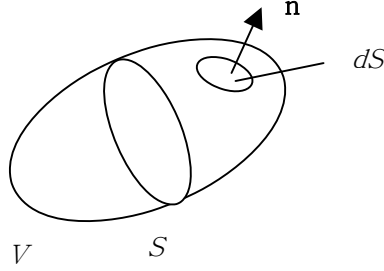


9S. Details on energy equation

[Derivation of energy equation]

1st law of thermodynamics: $e = q - w$ or $\dot{e} = \dot{q} - \dot{w}$



1st law of thermodynamics about the control volume displayed above:

Internal energy change = Heat flux + Work done by shear and normal (pressure) stress + work done by external force + work done by body force

$$\frac{d}{dt} \int_V \rho e dV = - \int_S \mathbf{q} \cdot \mathbf{n} dS + \int_S \sigma_{ij} v_i n_j dS + \int_S \mathbf{F}_{ext} \cdot \mathbf{v} dS + \int_V \rho \mathbf{F}_B \cdot \mathbf{v} dV,$$

where

$$\begin{aligned} \frac{d}{dt} \int_V \rho e dV &= \int_V \frac{\partial(\rho e)}{\partial t} dV + \int_S \rho e \mathbf{v} \cdot \mathbf{n} dS \quad \text{by transport theorem} \\ &= \int_V \left[\frac{\partial(\rho e)}{\partial t} + \nabla \cdot (\rho e \mathbf{v}) \right] dV \quad \text{by divergence theorem} \end{aligned}$$

$$\int_S \mathbf{q} \cdot \mathbf{n} dS = \int_V \nabla \cdot \mathbf{q} dV \quad \text{by divergence theorem}$$

$$\int_S \sigma_{ij} v_i n_j dS = \int_V \frac{\partial(\sigma_{ij} v_i)}{\partial x_j} dV \quad \text{by divergence theorem}$$

Therefore,

$$\int_V \left[\frac{\partial(\rho e)}{\partial t} + \nabla \cdot (\rho e \mathbf{v}) \right] dV = \int_V \left[- \nabla \cdot \mathbf{q} - \frac{\partial(\sigma_{ij} v_i)}{\partial x_j} + \rho \mathbf{F}_B \cdot \mathbf{v} \right] dV$$

For the above equation to be valid for an arbitrary control volume,

$$\frac{\partial(\rho e)}{\partial t} + \nabla \cdot (\rho e \mathbf{v}) = - \nabla \cdot \mathbf{q} - \frac{\partial(\sigma_{ij} v_i)}{\partial x_j} + \rho \mathbf{F}_B \cdot \mathbf{v},$$

where

$$\frac{\partial(\rho e)}{\partial t} + \nabla \cdot (\rho e \mathbf{v}) = \rho \frac{\partial e}{\partial t} + e \frac{\partial \rho}{\partial t} + e \frac{\partial(\rho v_i)}{\partial x_i} + \rho v_i \frac{\partial e}{\partial x_i} = \rho \frac{\partial e}{\partial t} + e \frac{\partial \rho}{\partial t} + e \nabla \cdot (\rho \mathbf{v}) + \rho \mathbf{v} \cdot \nabla e$$

Substituting the continuity equation (mass conservation), $\frac{\partial \rho}{\partial t} + \nabla \cdot (\rho \mathbf{v}) = 0$

$$\frac{\partial(\rho e)}{\partial t} + \nabla \cdot (\rho e \mathbf{v}) = \rho \frac{\partial e}{\partial t} + \rho v_i \frac{\partial e}{\partial x_i} = \rho \frac{\partial e}{\partial t} + \rho \mathbf{v} \cdot \nabla e = \rho \frac{de}{dt}$$

Consequently,

$$\rho \frac{de}{dt} = -\frac{\partial q_i}{\partial x_i} - \frac{\partial(\sigma_{ij} v_i)}{\partial x_j} + \rho F_{Bi} v_i \quad (\text{A})$$

Now, consider momentum equation

$$\rho \frac{dv_i}{dt} = -\frac{\partial(\sigma_{ij})}{\partial x_j} + \rho F_{Bi}$$

Taking the dot product with $\mathbf{v}$,

$$\rho v_i \frac{dv_i}{dt} = -v_i \frac{\partial(\sigma_{ij})}{\partial x_j} + \rho F_{Bi} v_i$$

$$\rho \frac{d\left(\frac{1}{2} v_i^2\right)}{dt} = -v_i \frac{\partial(\sigma_{ij})}{\partial x_j} + \rho F_{Bi} v_i \quad (\text{B})$$

Remembering $e = u + \frac{1}{2} v_i^2 + p.e.$ and neglecting the potential energy, (A)-(B) yields

$$\rho \frac{du}{dt} = -\frac{\partial q_i}{\partial x_i} + \sigma_{ij} \frac{\partial v_i}{\partial x_j},$$

where

$$\sigma_{ij} \frac{\partial v_i}{\partial x_j} = \left[-p \delta_{ij} + \tau_{ij} \right] \frac{\partial v_i}{\partial x_j} = -p \frac{\partial v_i}{\partial x_i} + \tau_{ij} \frac{\partial v_i}{\partial x_j}$$

Accordingly,

$$\rho \left[\frac{du}{dt} + \frac{p}{\rho} \frac{\partial v_i}{\partial x_i} \right] = -\frac{\partial q_i}{\partial x_i} + \Phi \quad (\Phi \equiv \tau_{ij} \frac{\partial v_i}{\partial x_j} : \text{viscous dissipation term})$$

Substituting the continuity relation $\frac{\partial v_i}{\partial x_i} = \nabla \cdot \mathbf{v} = -\frac{1}{\rho} \frac{d\rho}{dt} = \rho \frac{d(1/\rho)}{dt}$ and the definition of

enthalpy $h = u + pv = u + p/\rho$,

$$\rho \left[\frac{du}{dt} + \frac{p}{\rho} \frac{\partial v_i}{\partial x_i} \right] = \rho \left[\frac{du}{dt} + p \frac{d(1/\rho)}{dt} \right] = \rho \left[\frac{dh}{dt} - \frac{1}{\rho} \frac{dp}{dt} \right]$$

Assuming an isobaric process with negligible viscous dissipation (low velocity), i.e., $\frac{dp}{dt} \approx 0$,

$\Phi \approx 0$,

$$\rho \frac{dh}{dt} = \rho c_p \frac{dT}{dt} = -\nabla \cdot \mathbf{q}$$

Employing Fourier's law, $\rho c_p \frac{dT}{dt} = \nabla \cdot (k \nabla T)$

In Eulerian description, $\boxed{\rho c_p \left(\frac{\partial T}{\partial t} + \mathbf{v} \cdot \nabla T \right) = \nabla \cdot (k \nabla T)}$